## Topic 8: Origins of Genetic Variation

This topic builds on the knowledge of meiosis and natural selection. It considers the dihybrid inheritance of alleles and genes. The inheritance of unlinked and linked genes is studied. The effect of selection pressures on the allele frequencies in gene pools and their impact on speciation are discussed.

Opportunities for developing mathematical skills within this topic include: using ratios, fractions and percentages; understanding simple probability; using Chi squared tests; understanding measures of dispersion, including standard deviation and range; solving algebraic equations; the analysis of allele frequencies using the Hardy-Weinberg equation. (Please see *Appendix 6: Mathematical skills and exemplifications* for further information.)

| **Students should:** | |
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| **8.1** | **Origins of genetic variation** |
| i | Understand that mutations are the source of new variations and that the processes of random assortment and crossing over during meiosis give rise to new combinations of alleles in gametes. |
| ii | Understand how random fertilisation during sexual reproduction brings about genetic variation. |